

Fall 2025 EE582 Homework 02 Report

Nodal-Analysis-based Modeling of Electric Circuits

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Section A (Numerical Oscillations of Discretization Rules)

One way to analyze the properties of various discretization methods (e.g., in terms of their numerical oscillations) is to solve a lossless first-order system (e.g., inductance or capacitance) when the state variable of the component is suddenly changed. The circuit on the right considers a case where a dc voltage is suddenly applied to a capacitor. Because the source voltage is applied like a step function, we know that the current of the capacitor should look like an impulse, analytically. Let us assume that in the simulations all the variables are zero at $t = 0$ and the voltage source takes its value at $t = \Delta t$.

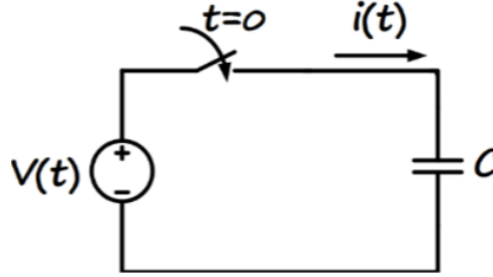


Figure 1: Simple capacitor circuit with DC voltage source

1) Determine the step-by-step solution for the current in the capacitor (in terms of capacitance C , time-step size Δt , and source voltage) using the following discretization rules: a) Forward Euler, b) Backward Euler, c) Trapezoidal.

Trapezoidal Discretization of Capacitor

The final equations for $v(t)$ and $i(t)$ are as follows:

$$\boxed{v(t) = e_h + R_C i(t)} \quad (1)$$

$$\boxed{i(t) = \frac{v(t)}{R_C} - i_h} \quad (2)$$

where:

- $R_C = \frac{\Delta t}{2C}$ is the equivalent resistance.
- $e_h = v(t - \Delta t) + R_C i(t - \Delta t)$ is the history voltage.
- $i_h = \frac{v(t - \Delta t)}{R_C} - i(t - \Delta t)$ is the history current.

The derivation of these equations can be found in appendix A.

Section A: Question 2

Comment on how the simulation time-step size impacts the numerical solution, and which method is preferred in this case.

Section B: Question 1

The parameters of the circuit below are: $R = 10\Omega$, $L = 20mH$, $e(t) = 10Vdc$, and the switch is closed at $t = 0$. The circuit is to be solved from $t = 0$ to $t = 10ms$ using:

- a) PSCAD software,
- b) your own MATLAB code based on the nodal analysis method.

A Derivation of Trapezoidal Discretization Equations

The governing equation for a capacitor is given by:

$$i(t) = C \frac{dv(t)}{dt} \quad (3)$$

Using the trapezoidal rule, the discretized forms for the voltage $v(t)$ and current $i(t)$ are derived as follows:

Core Relation:

$$v(t) - v(t - \Delta t) = \frac{\Delta t}{2C} [i(t) + i(t - \Delta t)] \quad (4)$$

Define the equivalent resistance R_C and history terms:

$$R_C = \frac{\Delta t}{2C} \quad (5)$$

1) Thevenin (Series) Companion

Rewriting the core relation:

$$v(t) = [v(t - \Delta t) + R_C i(t - \Delta t)] + R_C i(t) \quad (6)$$

Here, the capacitor at step t is a voltage source e_h in series with R_C :

$$v(t) = e_h + R_C i(t) \quad (7)$$

with the history voltage:

$$e_h = v(t - \Delta t) + R_C i(t - \Delta t) \quad (8)$$

2) Norton (Parallel) Companion

From the core relation:

$$\frac{v(t) - v(t - \Delta t)}{R_C} = i(t) + i(t - \Delta t) \quad (9)$$

Rewriting:

$$i(t) = \frac{v(t)}{R_C} - \left[\frac{v(t - \Delta t)}{R_C} - i(t - \Delta t) \right] \quad (10)$$

Here, the capacitor at step t is a current source i_h in parallel with conductance $1/R_C$:

$$i(t) = \frac{v(t)}{R_C} - i_h \quad (11)$$

with the history current:

$$i_h = \frac{v(t - \Delta t)}{R_C} - i(t - \Delta t) \quad (12)$$